

Particulate Matter Monitoring & Health Effects

Issue 2026-08-08 | Entry window 8 August 2026 | 14 records in scope

14RECORDS IN SCOPE
(13 REJECTED)**7**SENSING / EXPOSURE
RECORDS (50%)**7**SUBTOPIC CLUSTERS
REPRESENTED**2**TIER-A
RECORDS**5**EFFECT ESTIMATES
EXTRACTED

Signal of the day

Two records on opposite ends of the evidence hierarchy make the same argument: the particle-mass concentration a monitor reports is conditional on a co-exposure the monitor does not measure.

Hur et al. (*Environ Pollut*) pooled **262,288** deaths across eight Kyushu cities over 26 pollen seasons and fitted an explicit SPM × pollen interaction in a two-stage time-series. The marginal association of suspended particulate matter with respiratory mortality at lag 1 is barely distinguishable from null (**RR 1.010, 95% CI 1.000–1.019**), but it separates cleanly on pollen load: **1.021 (1.005–1.037)** above the 75th percentile against **1.003 (0.987–1.018)** below it, and the split is robust to the choice of cutoff. A mass-only instrument in Fukuoka in March therefore returns a number whose health meaning changes by a factor of seven depending on a biogenic co-pollutant it is blind to. Çulpan et al. (*Rev Environ Health*) reach the same conclusion by subtraction across **21 studies and 990,157** children: the fine fraction is the only exposure with a consistent blood-pressure association (**+0.59 mmHg systolic per 10 µg/m³, 0.28–0.90**), the coarse-fraction estimates do not survive sensitivity analysis, and NO₂ is null — but the authors' own stated limitation is that they could not fit multipollutant models because the primary studies did not report the exposure covariance. Both papers are asking the measurement side for the same thing: a co-exposure-resolved record, not a scalar.

What changed today

- **The thinnest window this watch has run: 14 records against 26 yesterday and 27 on 6 August.** This is a calendar artefact, not a signal — 8 August is a Saturday and PubMed's entry-date volume collapses at weekends. The PubMed connector returned **9 hits on the health axis and zero on the instrumentation axis**, the third consecutive empty sensing query from that leg, and the local harvester returned 4 and 0. The corpus is held up almost entirely by **Crossref by ISSN, which supplied 6 of the 7 sensing and exposure records**. Treat the daily counts for 8 August as a floor when they enter the weekly trend line.
- **Six of fourteen records — and six of seven on the sensing axis — carry no abstract anywhere.** All six are *Atmospheric Environment* / *Atmospheric Pollution Research* deposits made today; Crossref, Europe PMC and Semantic Scholar all return metadata only, and Semantic Scholar has no record of the DOIs at all. They are listed with design, metric and region but are **not summarised**, because a summary written from a title is a fabrication. The highest-value pickups are Madalipay et al. on provincial PM_{2.5} conversion factors for South Korea and Cuerda Barcaiztegui et al. on a two-year classroom ventilation and IAQ field study in Madrid.
- **Five effect estimates with intervals, down from sixteen yesterday**, and three of them are the same Kyushu interaction reported at different pollen strata. Two are co-pollutant (NO₂, indoor air nicotine). **No PM ratio estimate from a cohort entered the corpus today.** Çulpan et al.'s meta-analytic estimates are mean differences in mmHg and are deliberately excluded from the forest plot, which is a log-ratio axis — plotting 0.59 against a null of 1.0 would render a harmful effect as protective. They are carried in the digest text and the register instead.
- **The one high-temporal-resolution result is a co-pollutant one.** Shukla et al. (*BMJ Open*) fit a Poisson GAM/DLNM on *hourly* NO₂ against 18,943 adult asthma admissions in Birmingham and Solihull, and locate the association at a **3–6 hour lag** (RR 1.13, 1.01–1.26 at the 18 µg/m³ mean over lag 0–24 h, daytime). The lag structure is the transferable part: an exposure signal resolvable at 3–6 hours is invisible to a 24-hour filter-based method and is exactly the regime distributed low-cost sensing is built for. It is NO₂, not PM, so it enters as a methodological rather than a substantive record.
- **Nothing new on sensor calibration, drift or co-location.** Three Consensus sweeps (calibration and drift, co-location against reference monitors, satellite–surface fusion with humidity correction) returned 20 hits each, capped at 3 per sweep by the free tier, and **every returned DOI resolved to 2019–2025 or to a record already in seen.json**. OpenAlex returned HTTP 429 for the sixth consecutive run. The calibration literature was not silent today; the available discovery paths simply cannot see a single day of it.

Corpus analytics

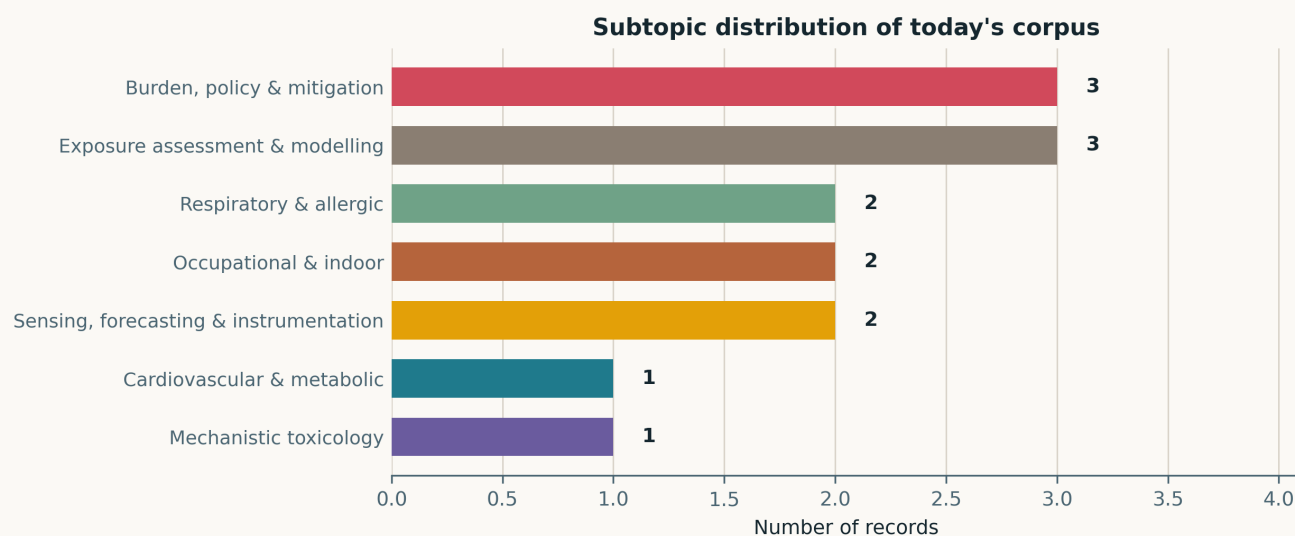


Fig. Subtopic assignment of the 14 in-scope records, single-label by dominant contribution. *Burden, policy & mitigation* and *Exposure assessment & modelling* tie at 3; adding *Sensing, forecasting & instrumentation* (2) and *Occupational & indoor* (2) puts **7 of 14 records on the measurement side**. Three of the ten canonical clusters are empty today — *Neuro / mental health*, *Reproductive & developmental* and *Other clinical endpoints* — which at $n = 14$ is expected sampling variation, not a field-level absence.

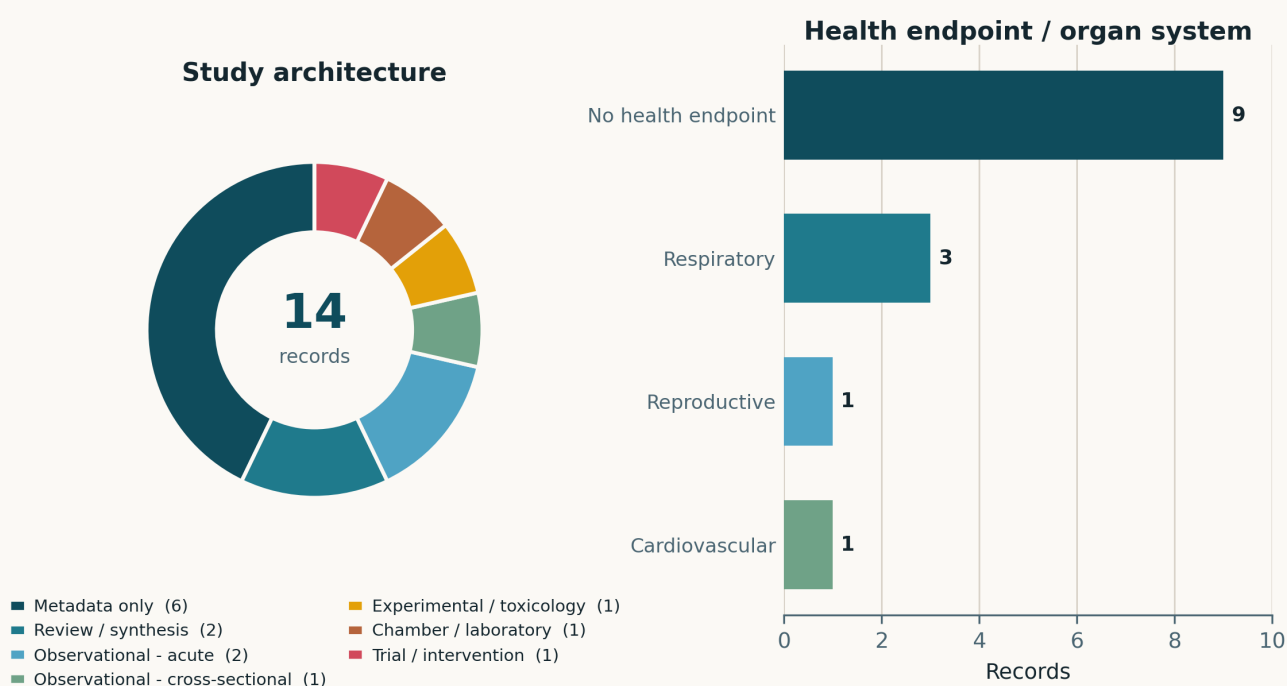


Fig. Left — study architecture. *Metadata only* is the single largest slice at 6 of 14 (43%), the highest share this watch has recorded, and it is an honest category rather than a residual: those records are listed but uncharacterised. No observational cohort entered today. Right — health endpoint. **Nine of fourteen records carry no human health endpoint** (six metadata-only sensing deposits, one control-technology bench study, one curriculum evaluation, one bibliometric review), so the endpoint panel describes five records and is not a field-level distribution.

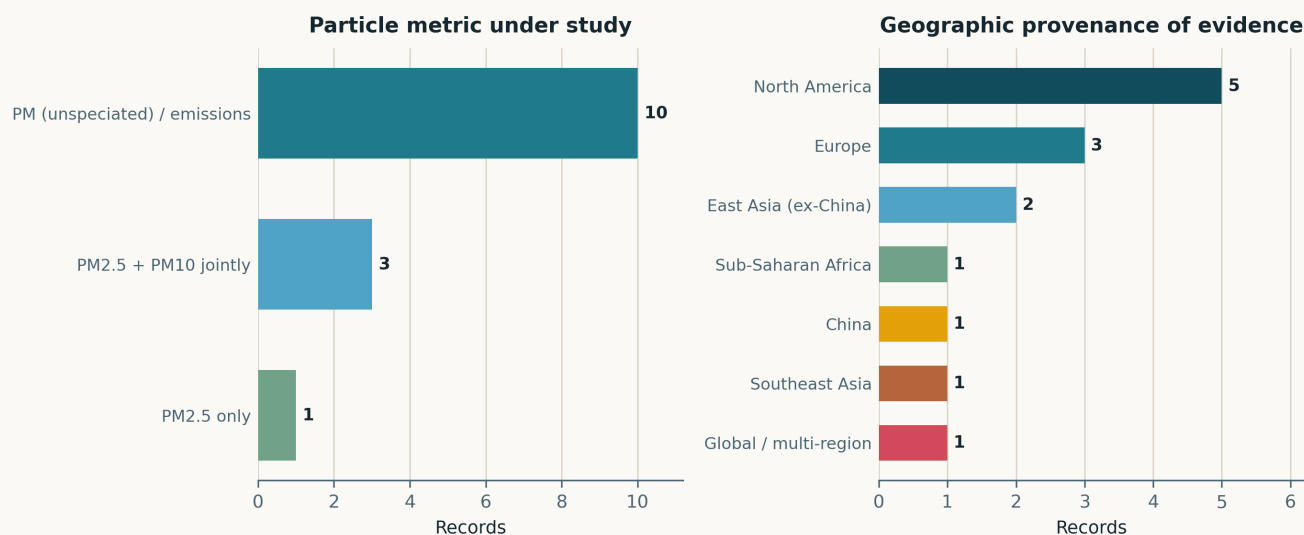


Fig. Left — particle metric. **Ten of fourteen records sit in PM (unspeciated) / emissions**, which is a direct consequence of the metadata-only share: without an abstract there is no basis for assigning a size cut. Only one record (Samburova et al., wildfire-influenced urban PM_{2.5} in northern Nevada) is unambiguously PM_{2.5}-only, and **no record today measures ultrafine particles** — a gap now standing at six consecutive issues. Right — geographic provenance. North America contributes 5; South Asia and Latin America are absent again.

Life-course windows addressed today (records may span several stages)

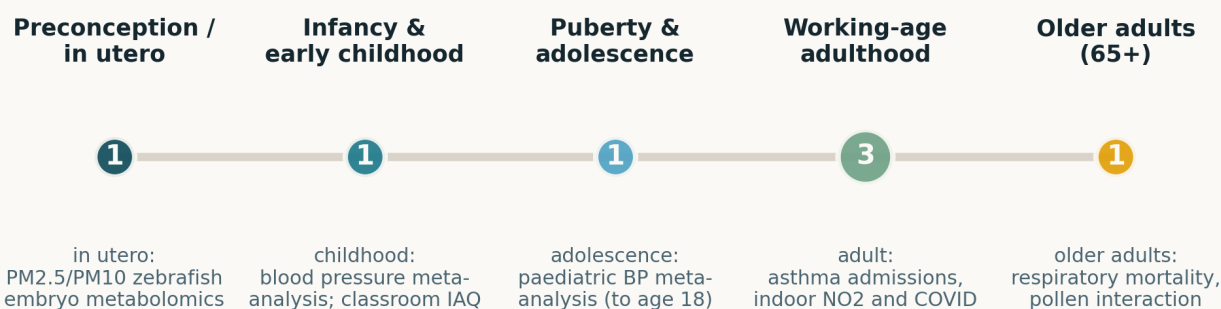


Fig. Life-course windows addressed. Counts are non-exclusive and total 7, not 14, because the six metadata-only records and the curriculum evaluation address no life stage. The paediatric blood-pressure meta-analysis spans childhood and adolescence and is counted in both.

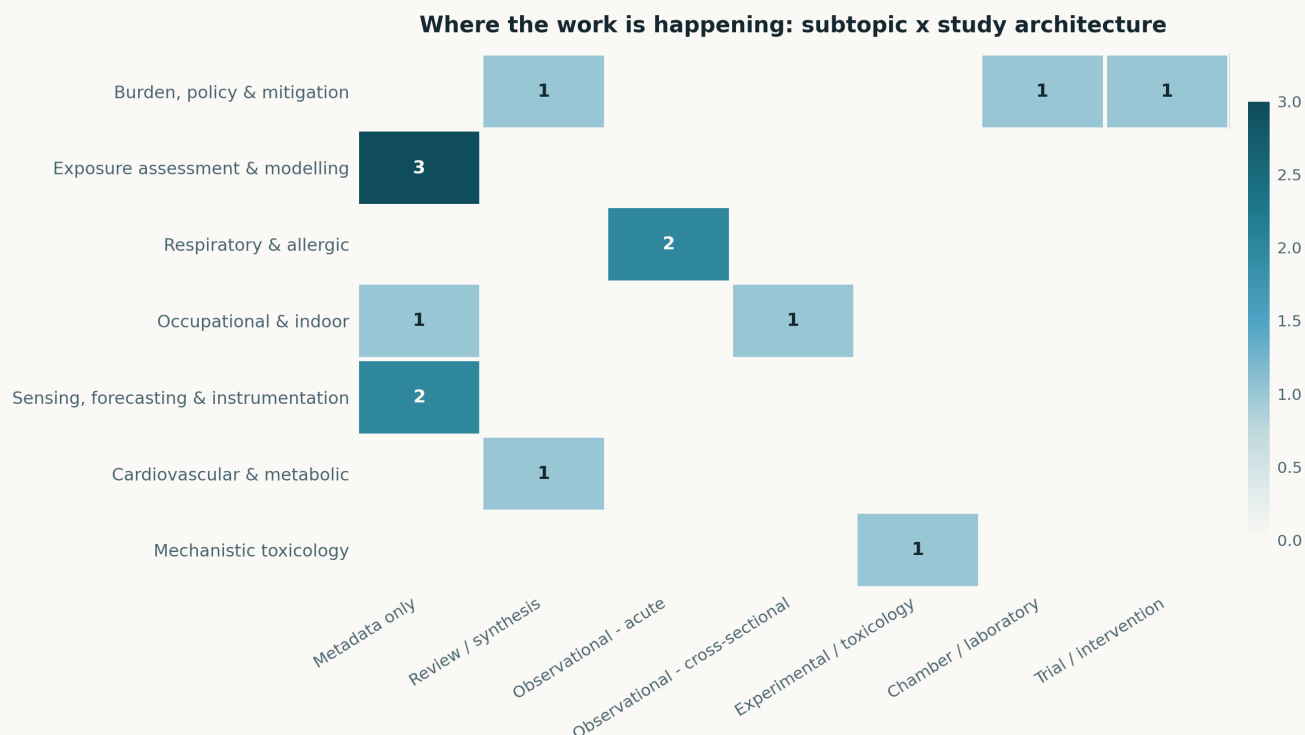


Fig. Subtopic × study-architecture cross-tabulation. The structural feature worth naming is that the entire *Metadata only* column (6 records) lands in the *Exposure assessment*, *Sensing* and *Occupational & indoor* rows — the abstract-availability failure is not randomly distributed across the corpus, it falls almost entirely on the instrumentation axis. This is the fourth issue in a row where the day's most methodologically relevant content is the least characterisable, and it is a property of Elsevier's atmospheric-sciences deposit pipeline, not of the literature.

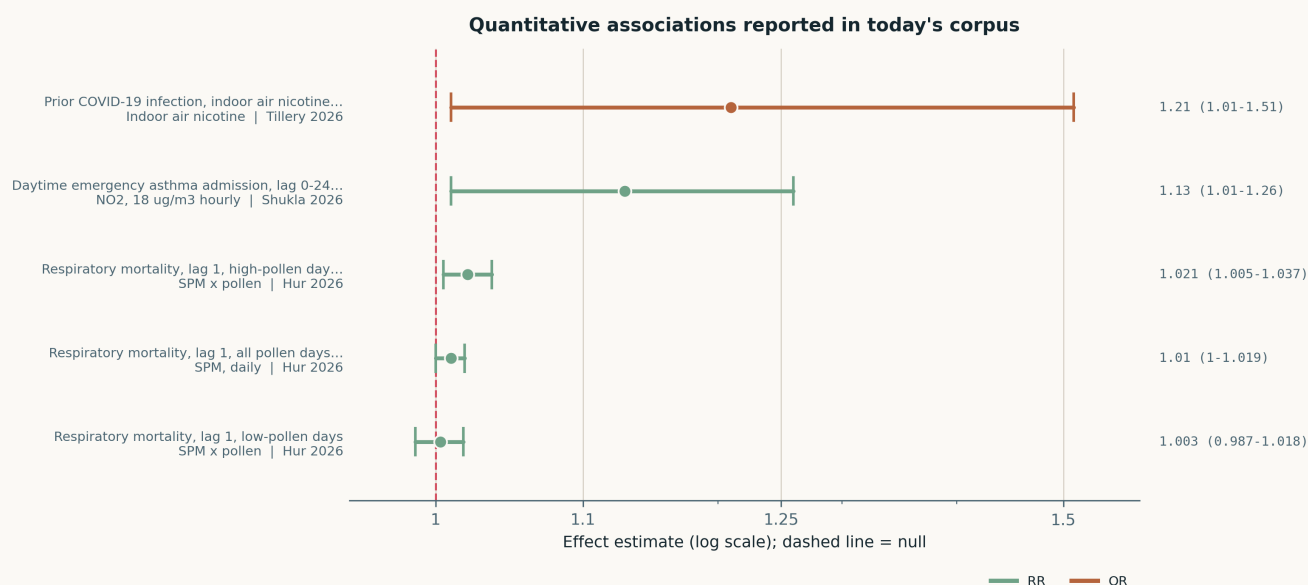


Fig. The five interval estimates in today's corpus, on a common logarithmic axis. Three of the five are the same Kyushu SPM–mortality association reported at different pollen strata and are therefore not independent; they are plotted together because the contrast between them is the finding. Two are co-pollutant rather than PM (Shukla, hourly NO₂; Tillery, indoor air nicotine). Exposure contrasts are heterogeneous — a daily SPM increment, an 18 µg/m³ hourly NO₂ mean, and a binary detection indicator — so the plot shows precision and direction only. Two estimates are deliberately absent: Çulpan et al.'s mmHg mean differences, which have no place on a ratio axis, and Tillery et al.'s long-COVID OR 3.69 (1.11–16.6), whose interval spans a factor of 15 and would compress every other estimate to a point (same exclusion rule applied to Jiang et al. on 7 August). Both are given in the digest text.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked METADATA ONLY were surfaced without an abstract in Crossref, Europe PMC or Semantic Scholar and are listed, not summarised.

Respiratory & allergic

2 records

Hur et al. 2026 Environ Pollut

Pollen modifies the SPM–mortality association across eight Kyushu cities

Two-stage time-series over 262,288 all-cause deaths, eight cities, pollen seasons (Feb–Apr) 1989–2014, with an explicit SPM × pollen interaction term in stage one and random-effects pooling in stage two. Respiratory mortality at lag 1: **RR 1.010 (1.000–1.019)** overall, **1.021 (1.005–1.037)** on days above the 75th-percentile pollen concentration against **1.003 (0.987–1.018)** below, stable across alternative cutoffs. The threat to inference is exposure misclassification in the modifier, not the exposure: pollen is measured at a single station per city and the interaction is fitted on a binary split of a highly skewed distribution, so the contrast is a lower bound on a continuous effect. Suspended particulate matter is a Japanese regulatory metric with a 100 % cut near 10 µm and is not interchangeable with PM_{2.5}. For distributed sensing the implication is concrete: an optical counter with a fluorescence or coarse-mode channel can resolve the modifier and the exposure on the same node, which no regulatory network in this study did.

doi: 10.1016/j.envpol.2026.128933

Shukla et al. 2026 BMJ Open

Hourly NO₂ and emergency admissions for acute asthma, Birmingham and Solihull

Poisson generalised additive model with a distributed-lag non-linear term on *hourly* counts — 18,943 adult asthma admissions to four acute hospitals, June 2016–May 2022, linked through the PIONEER data hub. The mean 18 µg/m³ NO₂ exposure over lag 0–24 h carried **RR 1.13 (1.01–1.26)** for daytime admissions, with the association concentrated at a **3–6 hour lag** and a steeper gradient in the most deprived quintile. Exposure is assigned from fixed monitors to residence, so the hourly resolution is nominal rather than personal — time-activity displacement over a 3–6 h window is precisely the scale at which residential assignment fails, which biases toward the null. The pollutant is NO₂, so this enters the watch on method rather than on substance: it is the cleanest recent demonstration that the health-relevant exposure signal lives below the 24-hour averaging period that filter-based reference methods impose.

doi: 10.1136/bmjopen-2025-110612

Cardiovascular & metabolic

1 record

Çulpan et al. 2026 Rev Environ Health

Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis

PRISMA-guided update over prior reviews plus PubMed and Web of Science to 27 March 2025; 21 studies, 990,157 participants in the qualitative synthesis, with meta-analysis restricted to the fine and coarse fractions and NO₂ because cohort data for SO₂, O₃ and CO were too sparse. **Fine PM: +0.59 mmHg systolic per 10 µg/m³ (0.28–0.90)**, the only estimate consistent across sensitivity analyses. The coarse fraction gave nominally tighter systolic (+0.28, 0.25–0.31) and diastolic (+0.27, 0.24–0.30) associations that *did not survive* sensitivity analysis — an implausibly narrow interval for a pooled estimate, which usually indicates a single dominant study. NO₂ was null for both. Several pollutants raised the odds of hypertension as a dichotomous outcome. Two limitations are load-bearing: exposure increments and hypertension definitions were not harmonised across primary studies, and no multipollutant model could be fitted because the primary studies did not report exposure covariance, so the pollutant-specific attribution is weaker than the point estimates suggest. Note also that the PubMed deposit strips subscripts from every particle-size label, so fraction assignment here is inferred from the stated meta-analytic scope.

doi: 10.1515/reveh-2026-0032

Occupational & indoor exposure

2 records

Tillery et al. 2026 Indoor Air

Indoor air nicotine, NO₂ and COVID-19 outcomes in a Great Plains tribal community

COVID-19 Wayakta He? study: 290 multigenerational Cheyenne River Sioux Tribe households, each pair of adults at least 9 years apart in age, with one week of passive air-nicotine and NO₂ badges plus saliva serology. Geometric mean air nicotine 0.052 µg/m³ (detected in 40.1 % of homes); geometric mean NO₂ 10.5 ppb, range 0.65–138, consistent with propane cooking and heating. **OR 1.21 (1.01–1.51)** for prior infection with detectable air nicotine; **OR 3.69 (1.11–16.6)** for self-reported long COVID with NO₂ exposure. The second interval spans a factor of fifteen and rests on a self-reported outcome in a cross-sectional design with one week of exposure measurement standing in for a multi-year window — it is a direction, not a magnitude. The study's real contribution is that it measured the microenvironment at all: no ambient network in the Northern Plains resolves household combustion, and community-deployed passive samplers are the only instrument that reached it.

doi: 10.1155/ina/7749725

Cuerda Barcaiztegui et al. 2026 Atmos Pollut Res

Classroom ventilation management and user activities vs. indoor air quality, Madrid primary schools

METADATA ONLY — no abstract in Crossref, Europe PMC or Semantic Scholar at the time of this issue. A two-year classroom field study on ventilation management and occupant activity is directly on this watch's deployment axis and is the single highest-priority abstract re-check for the next run.

doi: 10.1016/j.apr.2026.103170

Mechanistic toxicology

1 record

Yibcharoenporn et al. 2026 ACS Omega

Metabolic profiling of PM-induced developmental toxicity in zebrafish embryos

Zebrafish embryos exposed 4–50 hpf to fine and coarse PM at 500 and 1000 $\mu\text{g/mL}$, with untargeted LC-MS/MS metabolomics. The fine fraction alone produced a **22 h hatching delay** and disrupted pantothenate/CoA, retinol and riboflavin metabolism; riboflavin fell significantly at the higher dose. Glutathione responses diverged by size — reduced glutathione depleted under the fine fraction but *elevated* under the coarse — which is the most interesting result here because it argues against a single shared oxidative mechanism scaled by mass. The dose is the problem: 500–1000 $\mu\text{g/mL}$ of suspended PM is three to four orders of magnitude above any inhalable ambient concentration, so this establishes a mechanism, not a risk. The authors' closing suggestion of vitamin supplementation is not supported by anything in the design.

doi: 10.1021/acsomega.6c00972

Sensing, forecasting & instrumentation

2 records

Samburova et al. 2026 Atmos Pollut Res

Effects of western US wildfire smoke on urban air quality in northern Nevada

METADATA ONLY — no abstract available in Crossref, Europe PMC or Semantic Scholar. Long-range transported wildfire smoke over an urban receptor is the same problem class as Jin et al. (ACP, 7 August) on aged smoke in Missoula; worth a re-check once the publisher deposits the abstract.

doi: 10.1016/j.apr.2026.103168

Astaneh et al. 2026 Atmos Environ

Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions

METADATA ONLY. Using source-tagged particulate tracers as features in an ML bias-correction layer over a photochemical model is a transferable construction even though the target here is ozone, not PM.

doi: 10.1016/j.atmosenv.2026.122288

Exposure assessment & modelling

3 records

Madalipay et al. 2026 Atmos Pollut Res

Establishing PM_{2.5} conversion factors for South Korea at provincial level

METADATA ONLY. Provincial PM_{2.5} conversion factors bear directly on how a sparse regulatory network is extrapolated to unmonitored areas, which is the same inference a low-cost network is deployed to replace. Priority re-check alongside the Madrid classroom study.

doi: 10.1016/j.apr.2026.103171

McCerery et al. 2026 Atmos Environ

Arctic transport of North American and Canadian wildfire aerosols

METADATA ONLY. The only record in today's corpus with a polar or remote-marine receptor; long-range transport to a low-background site is where instrument detection limits, not calibration slope, become the binding constraint.

doi: 10.1016/j.atmosenv.2026.122289

Raptaki et al. 2026 Atmos Environ

Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation

METADATA ONLY. Enters on the secondary-organic-aerosol precursor link rather than on any direct PM measurement; tier C.

doi: 10.1016/j.atmosenv.2026.122283

Burden, policy & mitigation

3 records

Gu et al. 2026 J Hazard Mater

Electro-ultrasonic coupled agglomeration for removal of fine particles from biomass combustion

Segmented continuous-flow bench platform combining bipolar charging ($\pm 16\text{ kV}$), a 300 V/400 Hz AC field and 16 kHz/142 dB ultrasound, with particle-trajectory simulation and SEM. The coupled field reached **86.4 % light transmittance**, against single-field results 33.1 and 67.2 percentage points lower, with the extinction coefficient down to 0.9 m^{-1} ; simulated cumulative particle displacement rose >445 % as linear trajectories became three-dimensional helical paths, and SEM confirmed 15–20 μm agglomerates. The performance metric is the weakness: light transmittance and extinction coefficient are *optical* proxies, not gravimetric or number-resolved removal efficiency, so the headline “removal efficiency” is not directly comparable with any filtration literature that reports mass penetration. The submicron fraction claimed as the target is precisely the fraction an extinction measurement is least sensitive to.

doi: 10.1016/j.jhazmat.2026.143130

Wojcik et al. 2026 ATS Sch

An air pollution curriculum for pulmonary and critical care fellows

Kern six-step curriculum development across five domains — composition, health effects, *monitoring*, risk mitigation and environmental justice — with a 20-item knowledge scale built by modified Delphi and delivered as pre/post assessment over one academic year with spaced repetition. Knowledge rose in all five domains ($p < 0.001$). Single programme, no control arm, self-selected respondents and a knowledge rather than behaviour endpoint, so the effect size is uninterpretable; it is included because the explicit inclusion of *monitoring* as a clinical competency domain is a small but real indicator of where sensor-derived exposure data is expected to land clinically.

doi: 10.1093/atsscholar/aapag039

Kokol et al. 2026 Res Sq (preprint)

Artificial intelligence in the management of climate-change-related health issues in Africa

Scoping review over Scopus to October 2025, VOSviewer clustering a global against an African literature corpus. The divergence is the finding: global work clusters on environmental modelling, air-pollution mortality tracking and exposome integration, African work on supervised classification for infectious-disease triage and food security, with a “compute deficit”, data silos and power instability named as barriers. This is bibliometrics, not evidence — a VOSviewer map measures what Scopus indexed, and Scopus under-indexes African journals, so the gap is partly an artefact of the instrument. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-9910414/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Çulpan et al.	Long-term air pollution and blood pressure in children: updated systematic review and meta-analysis	<i>Rev Environ Health</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , NO ₂ (long-term, per 10 µg/m ³)	Global
Respiratory & allergic					
Hur et al.	Pollen modifies the association between suspended particulate matter and mortality across eight Kyushu cities	<i>Environ Pollut</i>	Time-series / case-crossover	SPM (daily, Japanese suspended PM)	Kyushu, Japan
Shukla et al.	Hourly ambient NO ₂ and emergency admissions for acute asthma, Birmingham and Solihull	<i>BMJ Open</i>	Distributed-lag time-series	NO ₂ (hourly, co-pollutant proxy)	Birmingham, UK
Mechanistic toxicology					
Yibcharoenporn et al.	Metabolic profiling of PM _{2.5} - and PM ₁₀ -induced developmental toxicity in zebrafish embryos	<i>ACS Omega</i>	Toxicological (in vitro)	PM _{2.5} + PM ₁₀ jointly (500, 1000 µg/mL suspensions)	Thailand (laboratory)
Occupational & indoor					
Cuerda Barcaiztegui et al.	Classroom ventilation management and user activities vs. indoor air quality in Madrid primary schools	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM (unspecified) + CO ₂ , indoor	Madrid, Spain
Tillery et al.	Indoor air nicotine, NO ₂ and COVID-19 outcomes in a Great Plains tribal community	<i>Indoor Air</i>	Cross-sectional survey	Indoor air nicotine + NO ₂ (1-week passive badges)	South Dakota, USA
Sensing, forecasting & instrumentation					
Astaneh et al.	Source-tagged PM tracers in random-forest bias correction of ozone design-value predictions	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified), source-tagged tracers	USA
Samburova et al.	Effects of western US wildfire smoke on urban air quality in northern Nevada	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} only (wildfire-influenced)	Nevada, USA
Exposure assessment & modelling					
Madalipay et al.	Establishing PM _{2.5} conversion factors for South Korea at provincial level	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} + PM ₁₀ jointly	South Korea
McCerery et al.	Arctic transport of North American and Canadian wildfire aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	PM (unspecified) wildfire aerosol	Arctic / North America
Raptaki et al.	Atmospheric BTEX in Heraklion, Crete: levels, sources and contribution to secondary aerosol formation	<i>Atmos Environ</i>	Metadata only (no abstract)	Secondary organic aerosol precursors (BTEX)	Heraklion, Greece
Burden, policy & mitigation					
Gu et al.	Electro-ultrasonic coupled agglomeration for removal of toxic fine particles from biomass combustion	<i>J Hazard Mater</i>	Chamber / laboratory	PM (unspecified) from biomass combustion	China
Kokol et al.	Artificial intelligence in the management of climate-change-related health issues in Africa: scoping review	<i>Res Sq (preprint)</i>	Scoping review	PM (unspecified), bibliometric corpus	Sub-Saharan Africa
Wojcik et al.	An air pollution curriculum for pulmonary and critical care fellows	<i>ATS Sch</i>	Educational intervention	PM (unspecified), curriculum content	USA

Provenance and method

Entry window 2026-08-08 to 2026-08-08, contiguous with the 7 August issue; no gap. Legs run: **PubMed** connector, two independent queries against [Date - Entry] — health axis 9 hits, instrumentation axis 0; a third MeSH-only sweep returned 10, of which 9 were water, soil or pharmaceutical-cleanroom monitoring and were rejected. The local harvester returned 4 health and 0 sensing. **Europe PMC** on CREATION_DATE returned 2 preprints, 1 carried. **Crossref by ISSN** returned 7 and is the only productive sensing leg, 6 carried. **Consensus** ×3 sweeps returned 20 each, capped at 3 per sweep by the free tier; all 9 returned records resolved to 2019–2025 DOIs or to `seen.json`, 0 carried. **arXiv** 50 records, none published or updated on or after 1 August. **OpenAlex** HTTP 429 (`from_created_date` leg remains broken). **13 records rejected**, each with a reason in `state/rejected.jsonl`. Six carried records have no abstract in any of the three checked sources and are listed without a summary; they are marked `METADATA ONLY` in the digest and in the architecture donut. Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. All DOI links resolve to the publisher of record. Content derived from PubMed and Crossref metadata; abstracts remain the copyright of their respective publishers.